



Maritime &
Coastguard
Agency

Guidance

MGN 657 (M+F) Amendment 2 Appendix

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APPENDIX MCA Fire Test Approval Procedures - Fixed Aerosol Fire Extinguishing Systems for Small Vessel Machinery Spaces

1. Definitions

1.1.1 **Aerosol** is a fire-extinguishing medium consisting of finely divided solid particles of chemicals released into a protected space as either condensed aerosol (pyrotechnic charge) or dispersed aerosol (compressed carrier gas).

1.1.2 **Generator** is a device for creating a condensed aerosol fire extinguishing medium by pyrotechnical means.

1.1.3 **Test Density** (g/m³) is the mass of a solid aerosol forming composition per cubic metre of the enclosure volume required to extinguish a specific type of fire as determined by the testing regime described below.

1.1.4 **Design Density** (g/m³) is the mass of a solid aerosol forming composition per cubic metre of the machinery space onboard provided to extinguish a specific type of fire, which is equal to the test density increased by safety factor of 1.3.

1.1.5 **Medium** – the aerosol product containing both the active fire suppressant and the by products of the combustion process which generated the aerosol stream.

1.1.6 **Agent** - medium for the purpose of these guidelines, these words are interchangeable.

1.1.7 **Casing** – The casing of the condensed aerosol generator, that is the metal outer part of the generator.

2. Test Enclosures

2.1 Figures (1) and (2) are for a small machinery space mock-up, constructed within a 20-foot ISO steel container. The container is one option; other enclosures may also be used for the test subject to approval by the Approved Body and with agreement of the MCA. A mock-up of an engine shall be created in the test enclosure to represent the expected fire scenarios more accurately than for an empty container with no obstructions to the seat of the fire. Figure (2) provides details for this mock up.

2.2 IMO MSC.1/Circular 1270 “Revised Guidelines for the Approval of Fixed Aerosol Fire-Extinguishing Systems for Machinery Spaces test methods” is used as the basis for these tests, with a smaller test enclosure volume and fire sizes to more accurately represent the small machinery spaces being approved.

2.3 The test enclosure volume should be at least equal to that of a standard 20-foot ISO container.

2.4 Tray A is placed in the enclosure away from obstructions including the mock engine. A simulated bilge system is created by plating-in close to the sides of the engine mockup, with a fuel tray (Tray B) placed underneath the mock ‘engine’ to simulate fuel accumulation from a leak. A fuel-spray is situated at the forward end of the engine mock-up, aiming across the engine and hidden by a plate placed over it. Also, in the four corners of the enclosure, small cans with 0.3 litres of heptane are placed at floor level in the four corners of the test enclosure to check the distribution of the extinguishing agent. The fire in all four cans shall be extinguished.

2.5 The Wood Crib should be made of kiln spruce and consist of four layers, each layer being made up of three members; the size of each member being 38 mm x 38 mm x 260 mm. Ignition of the crib is achieved by burning 1.5 litres of heptane on a 12.5 litre layer of water in a 500 mm x 500 mm x 100mm square pan with steel walls of 6mm thickness located underneath the crib. The bottom of the crib should be 140 mm above the floor. The members should be placed in 4 alternate layers at right angles to one another. Members should be evenly spaced forming a square structure. The pre-burn time for the wood crib is 2 minutes. The wood crib shall be located between trays A and B in Figure 1 below.

3. Required Fire Tests

3.1 A series of eight tests should be conducted as follows:

3.1.1 Open pool (Tray A) fire - diesel fuel (Diesel Fuel means light diesel or commercial fuel oil).

3.1.2 Hidden spray fire - diesel fuel.

3.1.3 Hidden pool fire (Tray B) - lubricating oil (Lube oil meaning 15W-40 heavy duty diesel engine oil, or similar).

3.1.4 Combined open pool (Tray A) / hidden spray - diesel fuel.

3.1.5 Combined open pool (Tray A) / hidden pool (Tray B) - diesel fuel / lube oil.

3.1.6 Combined hidden pool (Tray B) / hidden spray - lube oil / diesel fuel.

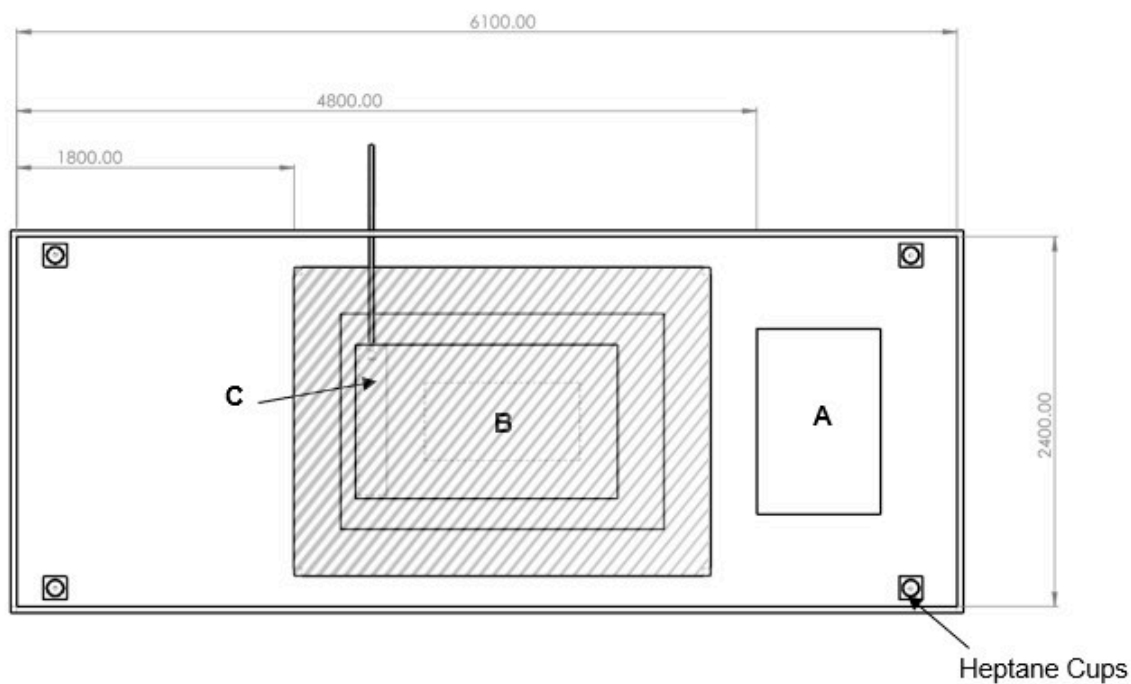
3.1.7 Combined open pool (Tray A) / hidden pool (Tray B) / hidden spray / heptane filled cans - diesel fuel / lube oil / diesel fuel / heptane. 3.1.8 Wood Crib (Class A fire)

Note: 3.1.1 to 3.1.7 are for Class B fires. Combined tests can be used to validate the effectiveness of single tests at the discretion of the approved body.

3.2 With reference to the Figures 1 and 2, Tray A should contain 10 litres of diesel fuel; Tray B should contain 5 litres of lubricating oil and 5 litres of diesel fuel. When required for the relevant fire test, trays should have a water base and may be started using heptane as an accelerant.

3.3 The hidden spray should provide a flow of 1 litre per minute at 3 bar pressure. The fuel spray should be shut-off 15 seconds after extinguishment. At the end of the 15 minute hold-time after the end of the aerosol discharge during which the test enclosure remains closed, the fuel spray should be restarted for 15 seconds prior to reopening the door and there should be no re-ignition.

Figure 1: Test enclosure (Plan View all Dimensions in millimetres)



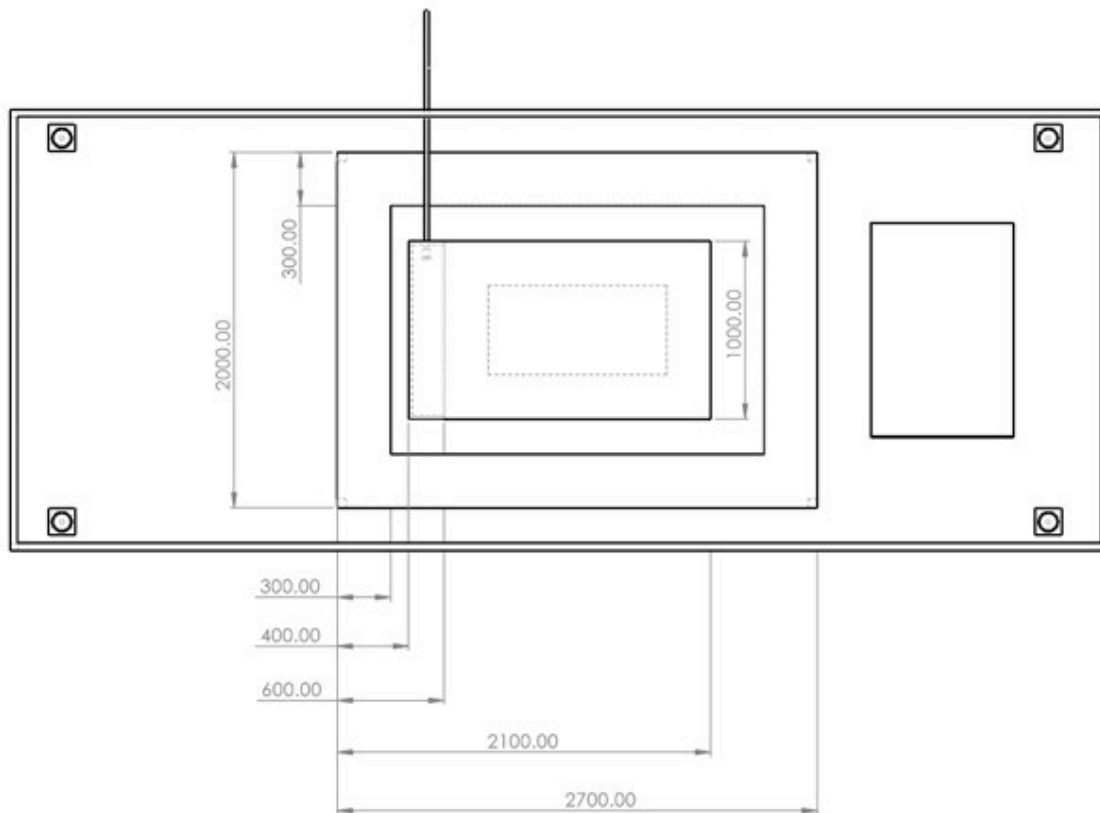
Shaded Area for Mock Engine Enclosure - Expanded in Figure 2]

Table 1: Fire scenarios

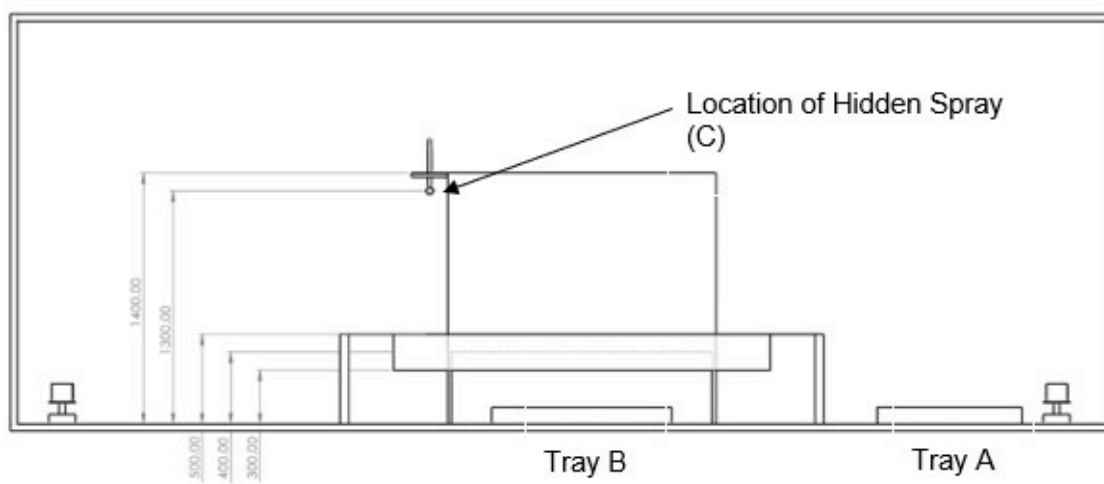
Ref.	Constituents	Quantity	Container type	Con
Tray A	Diesel fuel oil	10 L diesel fuel	Steel tray	1.20 = 0,9
Tray B	Engine lube oil and diesel fuel	5 L lube oil and 5 L diesel fuel	Steel tray	1.00 =0.5
Spray C	Hidden spray fire			
(Diesel fuel oil)	1 L/min	Pressurized container at 3 bar pressure	-	
Cans 1-4	Heptane	0.3 L/can	Open tin approx. 100 mm diameter	100 diam 100

Ref.	Constituents	Quantity	Container type	Con
Wood Crib	Kiln dried spruce or fir limber members. (Allowable moisture 9 % to13 %)	12 of 38 x 38 x 260 mm	-	The are t in 4 : layer right one : Mem be e spac a sq struc (spa 50 m wood be lc Tray place

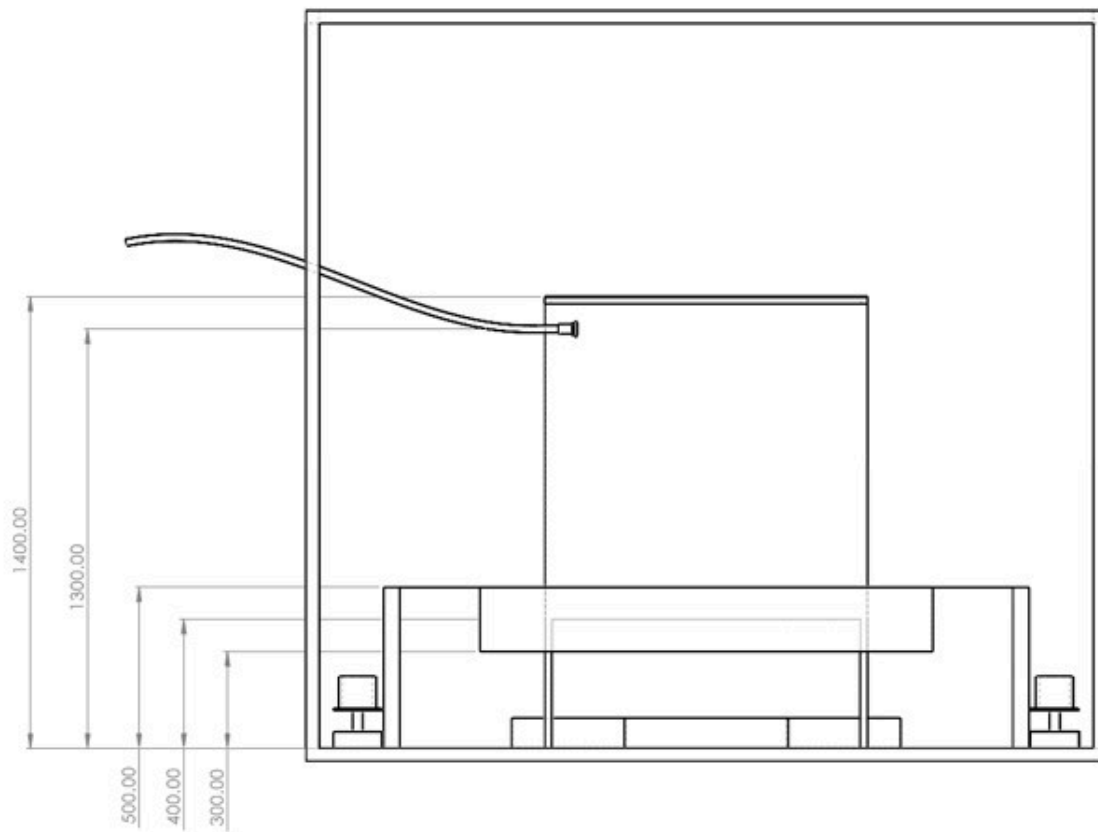
Figure 2: Engine mock-up (all Dimensions in millimetres)



Plan View



Elevation View



End View

3.4 A pre-burn time of 2 minutes for the tray fires, and 5 seconds for the spray fire should be allowed before the fire extinguishing system is activated. The doors to the test enclosures should be closed just prior to system release to ensure Oxygen levels are not reduced significantly pre-test. Times to extinguishment and re-ignition, if any, should be noted, together with other details.

3.5 A continuous measure at a sampling rate of at least 1 Hz should be made from temperature thermocouples located above each tray and spray fire. Pressure readings, flow rates and duration of the discharge time should also be recorded when dispersed aerosol systems are being tested. The extinguishing system should be installed in the test enclosure according to the manufacturer's design and installation instructions. The thermocouple should be located between 30 mm and 100 mm above the fire source. Additional thermocouples should be co-located at each generator unit in the discharge stream at a distance of 100 mm to 300 mm from the generator unit outlet. Measurement devices should be appropriately calibrated to a relevant standard agreed by the approved body with traceability accounted for and recorded.

3.6 An overall restriction is imposed to the effect that the system is only considered suitable for installation in the machinery spaces of small vessels of less than 24 metre load line length. The maximum height of the

protected space is 4m and the maximum floor area (including any areas covered by permanent installations like engines).

3.7 Please note that this test method has been agreed by the MCA for UK operation and may or may not be accepted by other maritime administrations.

3.8 The test arrangement should be representative of the expected arrangement in a ship's machinery space, for example the distance between generators and the seat of the fire should be similar.

3.9 The release of an extinguishing agent may produce significant over or under pressurization in the protected space. Measures to limit the induced pressures to acceptable limits may have to be provided.

4. Principal Requirements

4.1 The components to be tested should be supplied by the manufacturer together with design and installation criteria, operational instructions, drawings, technical data, certificate of conformity and manufacturing location sufficient for the identification of the components.

4.2 The test procedure is intended for the determination of the effectiveness of different aerosol agent extinguishing systems against Class A fires and Class B spray and pool fires.

4.3 For aerosol systems, the discharge time should not exceed 60 seconds to achieve 85 % of the design density. Systems may need to discharge in a shorter time for reasons other than for fire-extinguishing performance. This can be done by weighing of generator units before and after discharge to determine the quantity of aerosol released and by using the readings from the thermocouples co-located with the generator units to determine discharge start and end times.

4.4 The quantity of extinguishing agent for the protected space should be calculated at the minimum expected ambient temperature using the design density based on the net volume of the protected space.

4.5 The gross volume of a protected space is calculated from the dimensions of the space.

4.6 The net volume of the space is determined by removing the volume of the machinery in the space from the gross volume. The objects that occupy volume in the protected space include, but are not necessarily

limited to auxiliary machinery, boilers, condensers, evaporators, main engines, reduction gears, tanks and enclosed trunks.

4.7 When calculating the net volume of a protected space, the volumes of the bilges, or the fire extinguishing system itself should not be removed from the gross volume.

4.8 The volume of free air contained in air receivers that in the event of a fire may be released into the protected space also has to be considered and will add to the net volume.

4.8.1 The volume of air in air receivers is the volume of air that would be released in the event of a fire to the protected space ambient pressure and temperature. For example, a 10 litre air receiver at 10 bar pressure would have an equivalent volume of 100 litres when released to the protected space which is at a nominal 1 bar pressure.

4.9 Subsequent modifications to the protected space that alter the net volume of the space should require the quantity of extinguishing agent to be adjusted to meet the requirements.

4.10 No fire suppression system should be used which is carcinogenic, mutagenic or teratogenic at concentrations expected during use. The discharge of aerosol systems to extinguish a fire could create a hazard to personnel from the natural form of the aerosol, or from certain products of aerosol generation (including combustion products and trace gases from condensed aerosols). Other potential hazards that should be considered for individual systems are the following: noise from discharge, turbulence, cold temperature of vaporizing liquid, reduced visibility, potential toxicity, thermal hazard and potential toxicity from the aerosol generators, and eye irritation from direct contact with aerosol particles. Unnecessary exposure to aerosol media, even at concentrations below an adverse effect level, and to their decomposition products should be avoided. All aerosols used in fire-extinguishing systems should have non-ozone depleting characteristics.

4.11 All systems should employ two separate control actions for releasing the extinguishing medium into a protected space, such as lift cover and turn key or similar. Means should be provided for automatically giving audible warning of the release of fire-extinguishing medium into any space in which personnel normally work or to which they have access. The alarm should operate for a suitable period, (at least 20 seconds) before the medium is released. Exposure to aerosol media, even at concentrations below an adverse effect level, should be avoided, proper the use of the isolation switch will avoid inadvertent exposures.

4.12 Fixed aerosol fire extinguishing systems are only for use in not normally occupied spaces. However, IMO circular 1270 “Revised Guidelines for the Approval of Fixed Aerosol Fire-Extinguishing Systems Equivalent to Fixed Gas Fire-Extinguishing Systems, as referred to in SOLAS 74, for Machinery Spaces” (4 June 2008) contains further requirements for normally occupied spaces related to the No Observable Adverse Effect Levels (NOAEL) for aerosol systems and their by-products (such as Carbon Monoxide). Compliance with these requirements does not need to be demonstrated for systems to be used in not normally occupied spaces, however as the spaces are occasionally occupied, the MCA will review manufacturer’s evidence of aerosol and by-products toxicity before approving any systems for use in not normally occupied spaces.

4.13 In no case should a dispersed aerosol system be used with halocarbon carrier gas concentrations above the “Lowest Observed Adverse Effect Level (LOAEL) nor the “Approximate Lethal Concentration (ALC)”. Nor should a dispersed aerosol system be used with an inert gas carrier at gas concentrations above 52 % calculated on the net volume of the protected space at the maximum expected ambient temperature, without the use of controls.

4.14 The system and its components should be suitably designed to withstand ambient temperature changes, vibration, humidity, shock, impact, clogging, electromagnetic compatibility and corrosion normally encountered in machinery spaces. Generators in condensed aerosol systems should be designed to prevent self-activation at a temperature below 250 °C. This can be demonstrated through existing test reports or through new testing to the satisfaction of the approved body.

4.15 The system and its components should be designed, manufactured and installed in accordance with standards acceptable to the International Maritime Organisation (IMO) and the National Authority (MCA). As a minimum, the design and installation standards should cover the following elements:

4.15.1 Safety:

- Toxicity.
- Noise - Generator/Nozzle discharge.
- Decomposition products.
- Obscuration.

4.15.2 Storage container design and arrangement strength requirements:

- Maximum / minimum fill density, operating, temperature range.

- Pressure & weight indication.
- Pressure relief.
- Agent identification, production date, installation date & hazard classification.

4.15.3 Agent supply, quantity, quality standards, shelf life & service life of an agent and igniter.

4.15.4 Handling & disposal of generator after service life.

4.15.5 Pipes and fittings:

- Strength, material properties, fire resistance and cleaning requirements.

4.15.6 Valves:

- Testing requirements; and
- Elastomer compatibility.

4.15.7 Generators/Nozzles:

- Height & testing requirements; and
- Elevated temperature resistance.

4.15.8 Actuation & Control Systems:

- Testing requirements; and
- Backup power requirements.

4.15.9 Alarms & Indicators:

- Predischage alarm, agent discharge alarms & time delays.
- Supervisory Circuit requirements.
- Warning signs, audible & visual alarms.
- Annunciation of faults (fault warning).

4.15.10 Enclosure integrity & leakage requirements:

- Enclosure leakage.
- Openings.
- Mechanical ventilation interlocks.

4.15.11 Design and test density requirements, total flooding quantity.

4.15.12 Agent flow calculation:

- Verification & approval of design calculation method.
- Fitting losses and/or equivalent length.

4.15.13 Inspection, maintenance, service and testing requirements.

4.15.14 Handling & storage requirements for pyrotechnical components.

4.16 Agent containers may be stored within a protected machinery space if the containers are distributed throughout the space. The arrangements of generators, containers, electrical circuits & piping essential for the release of any system should be such that in the event of damage to any one power release line through fire or explosion in the protected space (i.e. a single fault concept), at least the test density of the fire-extinguishing charge as required in paragraph 4.15.11 above can still be discharged having regard to the requirement for uniform distribution of medium throughout the space.

4.17 For all ships, the fire-extinguishing system design manual should address recommended procedures for the control of products of agent decomposition. The decomposition products should not be discharged in the vicinity of assembly spaces.

4.18 Spare parts and operating & maintenance instructions for the system should be provided as recommended by the manufacturer.

5. Success Criteria:

5.1 Class B fires should be extinguished within 30 s of the end of discharge. At the end of the hold period there should be no re-ignition upon opening the enclosure.

5.2 The fuel spray should be shut off 15 s after extinguishments. At the end of the hold time, the fuel spray should be restarted for 15 s prior to reopening the door and there should be no re-ignition.

5.3 The ends of the test fuel trays should contain sufficient fuel to cover the bottom of the tray.

5.4 The wood crib weight loss should be no more than 30% during the 2 min pre-burn test.

5.5 A re-ignition test should be conducted after the successful extinguishments of the tell-tale fires in test 1 (Fire A) within 30 s after completion of discharge. The test should involve the attempted ignition of two of the tell-tale fire containers. One container should be at the floor level and the other at the ceiling level at the diagonally opposite corner. At 10 min after extinguishment of the fires, a remotely operated electrical ignition source should be energized for at least 10 s at each container. The test should be repeated at 2 min intervals two more times, the last at 14 min after extinguishment. Sustained burning for 30 s or longer of any of these ignition attempts constitutes a re-ignition test failure.

6. Test Report:

6.1 The test report should include the following information:

6.1.1 Name and Address of the Test Laboratory.

6.1.2 Date and Identification Number of the Test Report.

6.1.3 Name and Address of Client.

6.1.4 Purpose of the Test.

6.1.5 Method of Sampling System Components.

6.1.6 Name and Address of Manufacturer or Supplier of the Product.

6.1.7 Name or other Identification Marks of the Product (for example serial numbers, date of manufacture, expiry date etc.).

6.1.8 Description of the Tested Product:

6.1.8.1 Drawings.

6.1.8.2 Descriptions.

6.1.8.3 Assembly Instructions.

6.1.8.4 Specification of included Materials.

6.1.8.5 Detailed Drawing of Test Set-up.

6.1.9 Date of Supply of the Product.

6.1.10 Date of Test.

6.1.11 Test method.

6.1.12 Drawing of each test configuration.

6.1.13 Identification of the Test Equipment and Instruments used.

6.1.14 Conclusions.

6.1.15 Deviations from the Test Method, if any.

6.1.16 Test Results including measurements and observations during and after the test.

6.1.17 Date, Name and Signature of qualified person witnessing the test.

7. Certificate:

7.1 On successful completion of the eight tests mentioned in paragraph 1.6 a Certificate of Inspection and Tests will be issued and the Aerosol system / product will be considered suitable for installation in normally unoccupied spaces containing fuel having a flash point of not less than 43 °C (closed cup test), of vessels of less than 24 metres load line length, where the space to be protected does not exceed a deck height of 4 metres, or deck area greater than 64 square metres equal to 256 m³, this volume is the maximum permitted volume for approvals under this MGN. The certificate will state that the system/product is acceptable for the purpose of complying with the requirements of:

7.1.1 MGN 280 - Small Vessels in Commercial Use for Sport or Pleasure, Workboats and Pilot Boats - Alternative Construction Standards. As Amended.

7.1.2 The Codes of Practice for the Safety of Small Commercial Motor or Sailing Vessels of up to 24 metres Load Line length. As amended.

7.1.3 "The Safety of Small Workboats and Pilot Boats - A Code of Practice. As amended.

7.1.4 MSN 1871 (F) The Code of Practice for the Safety of Small Fishing Vessels of less than 15m Length Overall. As amended.

7.1.5 MSN 1872 The Code of Safe Working Practice for the construction and use of fishing vessels of 15m length overall to less than 24m registered length. As amended

7.1.6 The Codes of Practice for Police Boats. As amended.

7.1.7 Non-Category A Machinery Spaces in domestic passenger ships of Class III to VI(A) where the requirement was introduced by the Merchant Shipping (Safety Standards for Passenger Ships on Domestic Voyages) (Miscellaneous amendments) Regulations 2022.

8. References:

(a) International Maritime Organization MSC.1/Circ.1270 of 4 June 2008

“Revised Guidelines for the Approval of Fixed Aerosol Fire Extinguishing Systems Equivalent to Fixed Gas Fire Extinguishing Systems, as referred to in SOLAS 74, for Machinery Spaces.

(b) International Maritime Organization MSC/Circ. 668 of 30 December 1994 “Alternative arrangements for Halon Fire Extinguishing Systems in Machinery Spaces and Pump Rooms”.

(c) IMO 2010 FTP Code. International Code for Application of Fire Test Procedures, 2010 ISBN 978 92 801 1548 2 (International Maritime Organization).



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